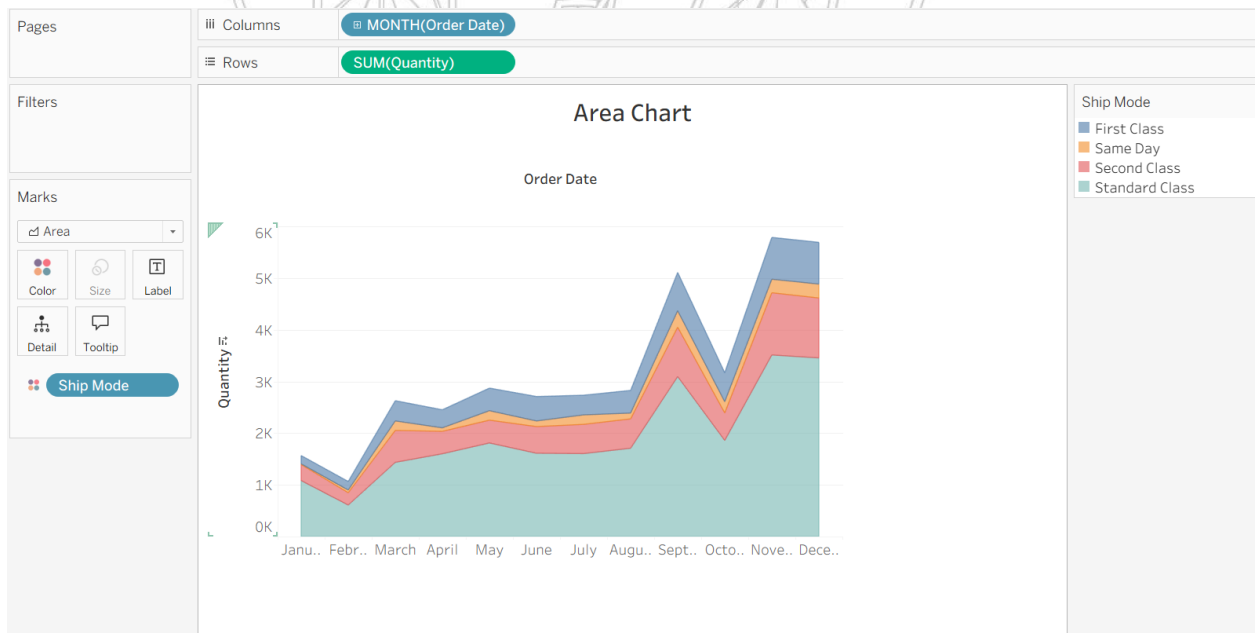


DATA ANALYTICS WITH TABLEAU ASSIGNMENT

Sheet-01: Area Chart

- There will be **data pane** in left side, drag **Order Date** and put it in the columns shelf.
- After right click on the **YEAR (ORDER DATE)** on the columns shelf.
- Then select **month**.
- In data pane, drag **QUANTITY** to the rows shelf.
- Drag the **Ship mode** to the marks card and add color on the marks card.
- On the **mark card**, select the drop-down menu and select the area the chart type.

OUTPUT:



Sheet- 02: Bar Chart

- To rows shelf, drag **sales** and place it.
 - In tableau sales will be **SUM (Sales)** automatically aggregates
 - Then **Order Date** is used, Tableau first shows a **LINE CHART** which is representing yearly sales trends.
- Click the drop-down menu on the **marks card**.
- Then select **Bar**.

Add the SHIP MODE COLORS

- Drag **ship mode** to the marks card.
- Drag the Region and place it in the Rows shelf on the left of sales.
- Filter only **WEST REGION**.
- Again drag the **region** to the filters shelf.

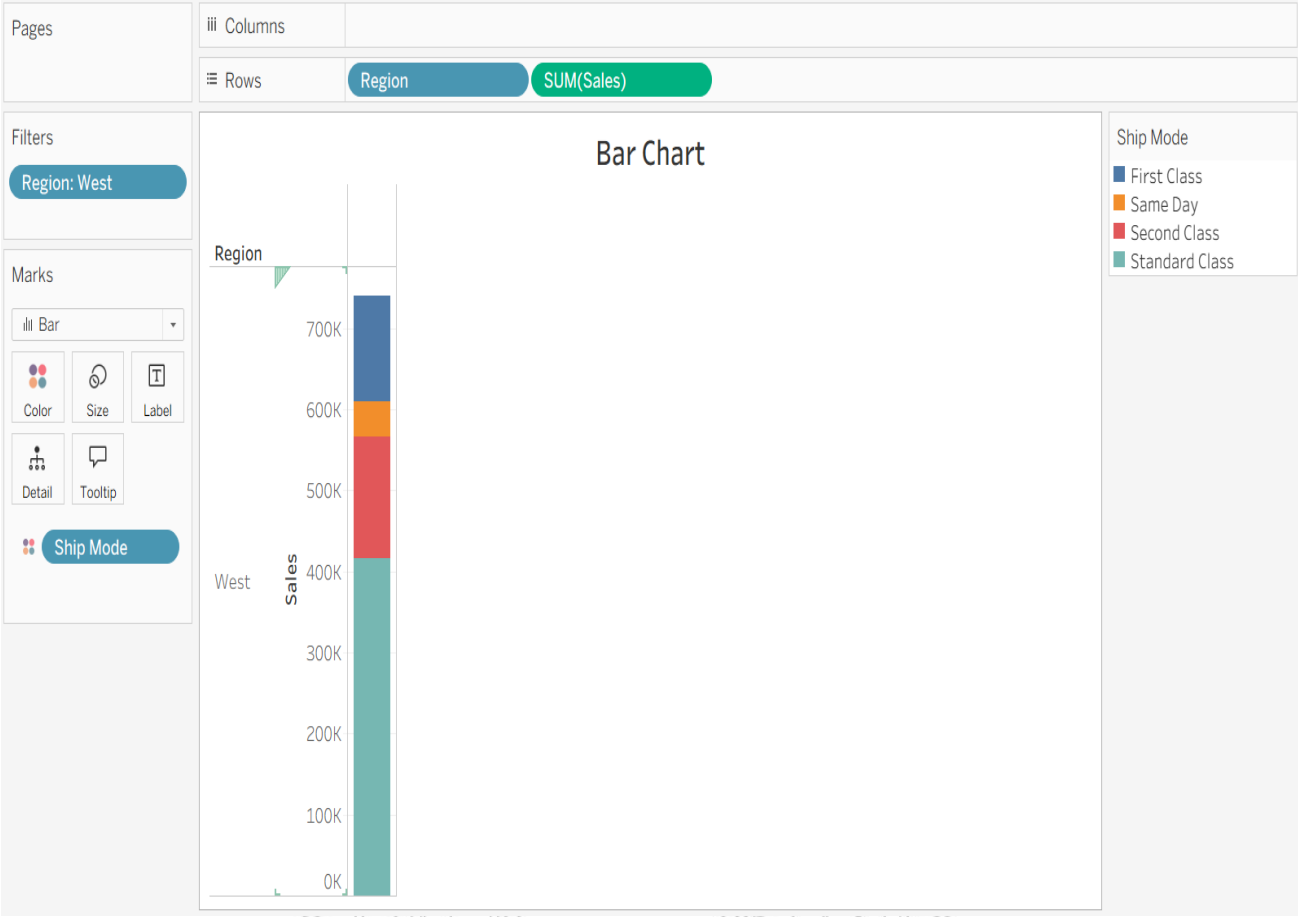
In filter dialog box:

- Uncheck **Central**
- Uncheck **East**
- Uncheck **South**
- Keep only **West**
- click ok [done]

OUTPUT:

BRAVO
- BUILDERS -
—





BRAVO

— BUILDERS —



Sheet – 03: Box Plot

- Drag the **segment**, place it in the **columns shelf**.
- Drag the **Discount**, place it in the **Rows shelf**.

Change Aggregation to Average

- On the Rows shelf, **click Discount**.
- Then hover over **Measure (Sum)** and change the aggregation to Average.

Adding Region

- Drag **region** to columns shelf. dropping it right of segment.

Covert to box plot

- click **show me** in toolbar.
- Then select **box-and-whisker plot**.

Now change columns shelf layout

- Drag **Region** from **Marks card** back to columns shelf, right of segment.

Disaggregate Date

- Now go to **analysis** and uncheck Aggregate Measure.

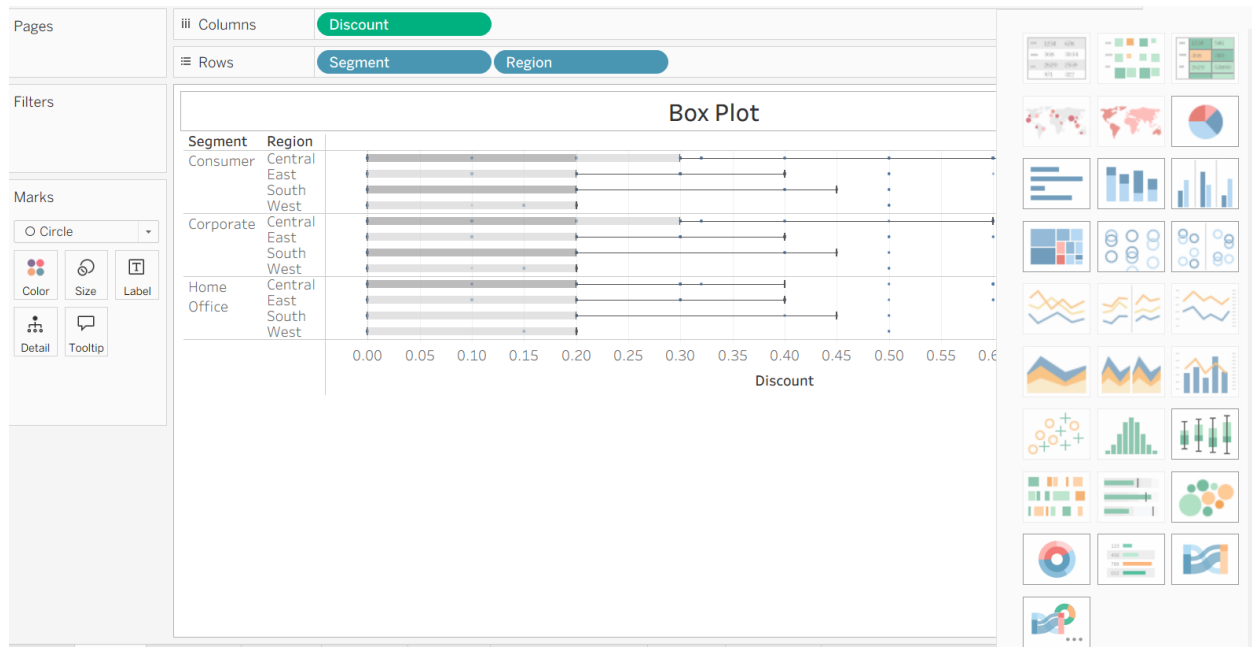
Swap Axes (Horizontal View)

- Now click swap rows and columns button (**Ctrl+ W**).

OUTPUT:

BRAVO
– BUILDERS –
—





BRAVO
— BUILDERS —



Sheet – 04: Gantt Chart

- Add **Order date** to columns.
- Tableau automatically shows it as **YEAR(Order Date)**.
- Now click on the **year** and change it to week number.
- Now it will be the **weekly** timeline chart.
- Drag **sub-category** to the rows shelf.
- Drag **ship mode** to rows shelf then place it right side of the sub-category

Now rows will shows as

-  Furniture first class Standard class etc.....
- In marks card select Gantt chart.
- Now small thin bars appear across timeline.
- Now next step, go to **analysis**.

Create calculated Field

- To create a calculated Field Name: **OrderUntilShip**.
- By using formula of **DATEDIFF('day', [Order Date],[Ship Date])**.
- And click okay or apply.

Add Delay to size

- Next, drag **OrderUntilShip** to size on the marks card.
- Now bars will become longer based on shipping days.

Change to sum to Avg

- on mark card right click on **SUM(OrderUntilShip)** and select measure to average.

Add colors by ship mode

- Now drag **ship mode** to color in the mark card.
- And now ship mode gets different color.

OUTPUT:



Sheet – 05: Heat Map

Adding Segment

- Drag **segment** to columns shelf.
- Now you will see.
 - Consumer
 - Corporate
 - home office

Adding Region and sub category to Rows

- Drag **Region** to rows shelf.
- Drag **sub-category** to row shelf and place it beside right of region.

Now hierarchy becomes:

- Central ----> Chairs, Tables, Phones etc.
- East
- South
- West

Add profit to color

- Drag profit and drop it in the marks card on color.

Now Tableau colors cells based on profit.

- High profit = Blue/ Dark
- Loss = Orange

Add Border to Cells

- Now adding border to cells by click color on **mark card**.
- Click down arrow and select Edit color.
- Then under **Border**, choose **medium gray** to make each box clearer.

Change the color palette

- On profit color legend click dropdown arrow.

- Select Edit color.

In the dialog box:

- chose palette ---->Red-black diverging.
- check Use full Color Range and click Apply on Okay.
- Now the profit = black/dark.
- And loss = red.

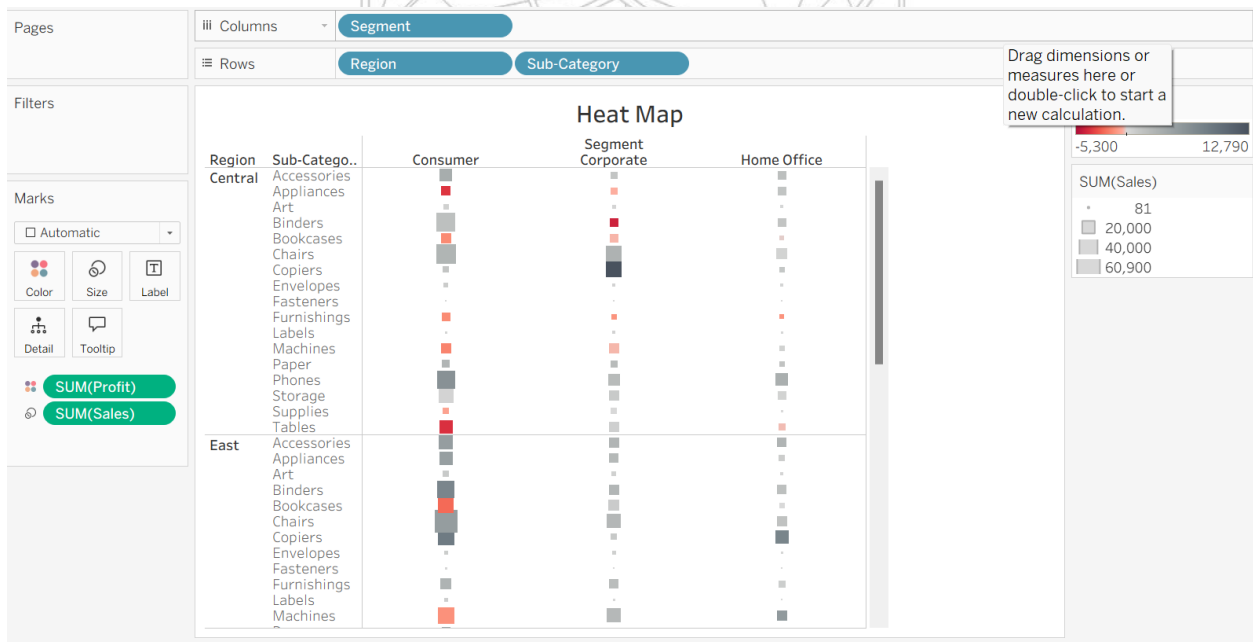
Add Sales to Size

- Drag **Sales** and drop it on the **SIZE** in the marks card.
- Now the box size represents sales amount.
 - Large box =high sales
 - Small box =low sales

Increase size

- click **size** on marks card and move it slider to right until the boxes.
- Fit properly in the **chart** to get output.

OUTPUT:



Sheet – 06: Histogram

- Drag **QUANTITY** and place it on columns shelf.

Create Histogram Using show me

- On top right corner, click **show me** and select Histogram.

Now tableaus will automatically does :

- On columns creates **Quantity (bin)**.
- **Quantity** will moves to Rows.
- **CNT (QUANTITY)** changes aggregation.

Understand Histogram

- Now **X-axis**: quantity.
- **Y-axis**: count of orders.

Example:

- Around 2400 Orders had quantity = 4
- Around 4800 Orders had quantity = 2

Add segment to color

- Drag **segment** from dimensions and drop it on color in **marks card**.

Now bars becomes stacked by show as

- Consumer
- corporate

- Home office

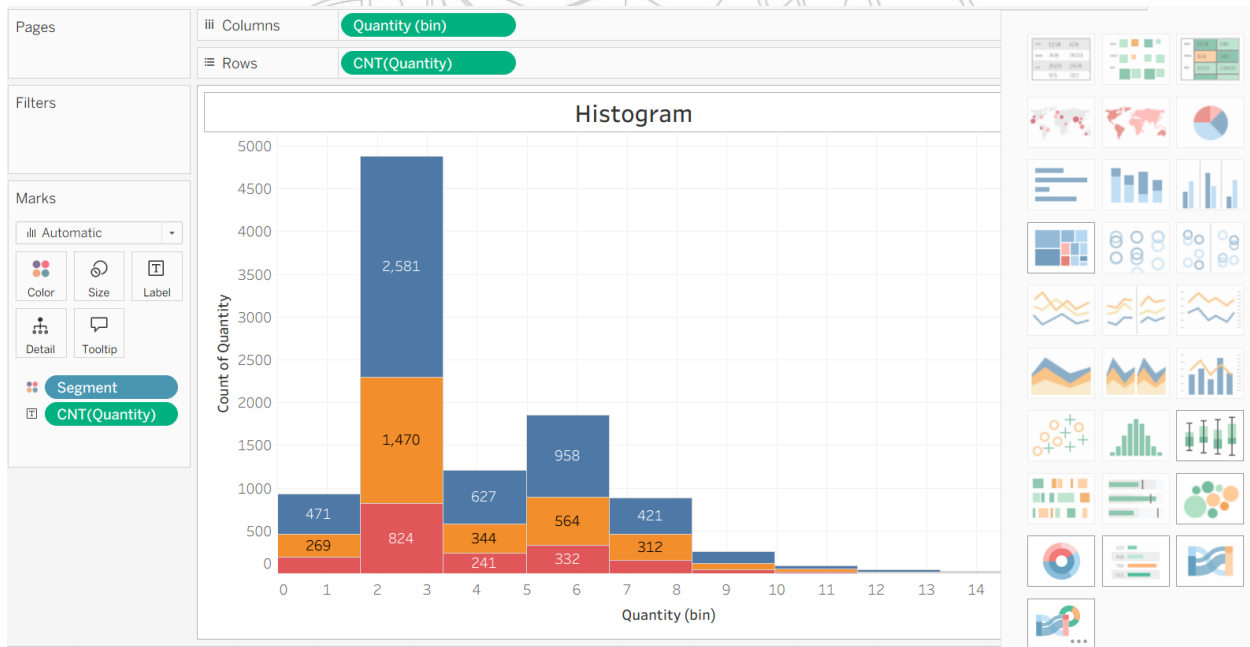
Copy Count to Label

- Drag CNT(Quantity) from rows shelf to label by using Ctrl.

Convert labels to percent of Total

- On mark card right click **CNT(Quantity)**.
- Then select **quick table calculation to percent of total** now labels show percentages.

Output:



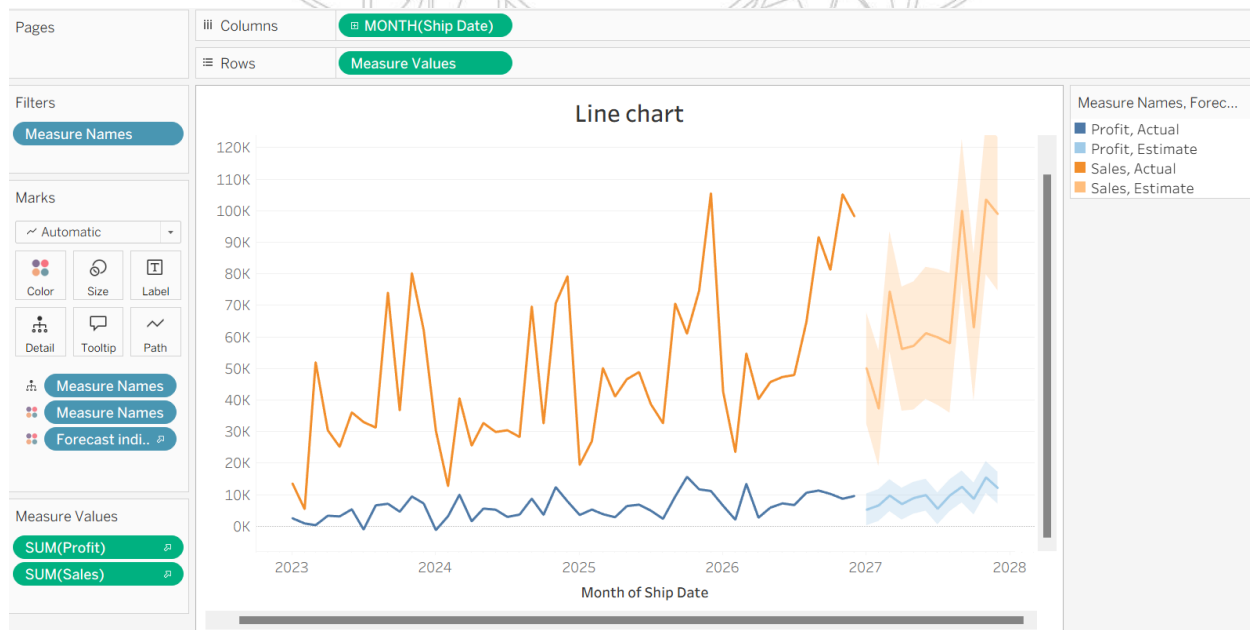
BRAVO
- BUILDERS -



Sheet – 07: Line Chart

- Drag **ship Date** and place it on **columns shelf**.
- Tableau shows: **YEAR (SHIP DATE)**.
- Now drag **sales** and place it on the **rows**.
- Then a **line chart** will be appear.
- Drag **profit** and drop it in the **rows shelf**, right side of the **sales**.
- Tableau creates two separate axes:
 - Sales axis
 - Profit axis
- Drag **SUM(Profit)** from Rows shelf.
- And then drop directly on the **sales axis**.
- Now **profit** and **sales** share **same axis**.
- Repeating yearly spikes Seasonal business growth And this is called **seasonality**.
- In columns shelf **click month** and change it to the **year**.
- Then go to **analysis pane** and add **forecast into chart area** and next change it back year to month.

OUTPUT:



Sheet – 08: Packed Bubbles

- Adding **category** to columns shelf.

Now, you will see:

- Furniture
- Office supplies
- Technology

Add Sales to Rows

- Drag **sales** and place it on the **Rows shelf**.
- **Bar chart** created by default.

Change to Packed Bubbles

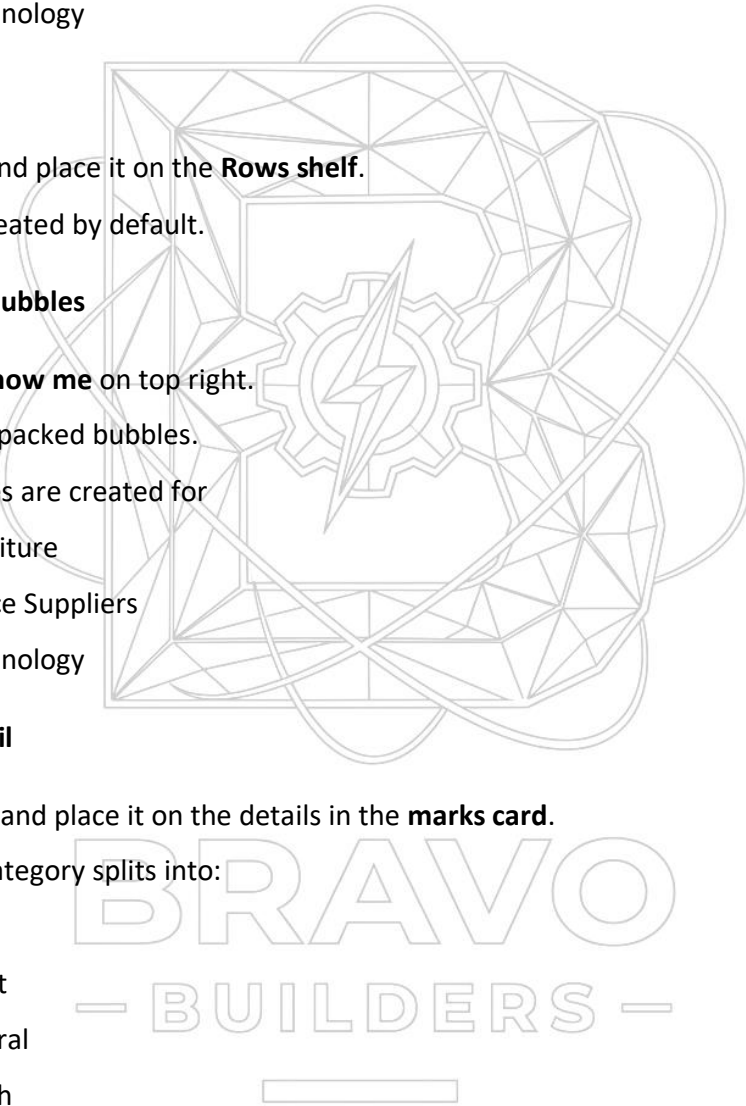
- Select the **show me** on top right.
- Then **select** packed bubbles.
- Then bubbles are created for
 - Furniture
 - Office Suppliers
 - Technology

Add Region to Detail

- Drag **region** and place it on the details in the **marks card**.
- Now each category splits into:
 - East
 - West
 - central
 - south
- Now more bubbles will appear.

Add profit to color

- Drag **profit** and place it in the **color**. Now bubble color represents profit



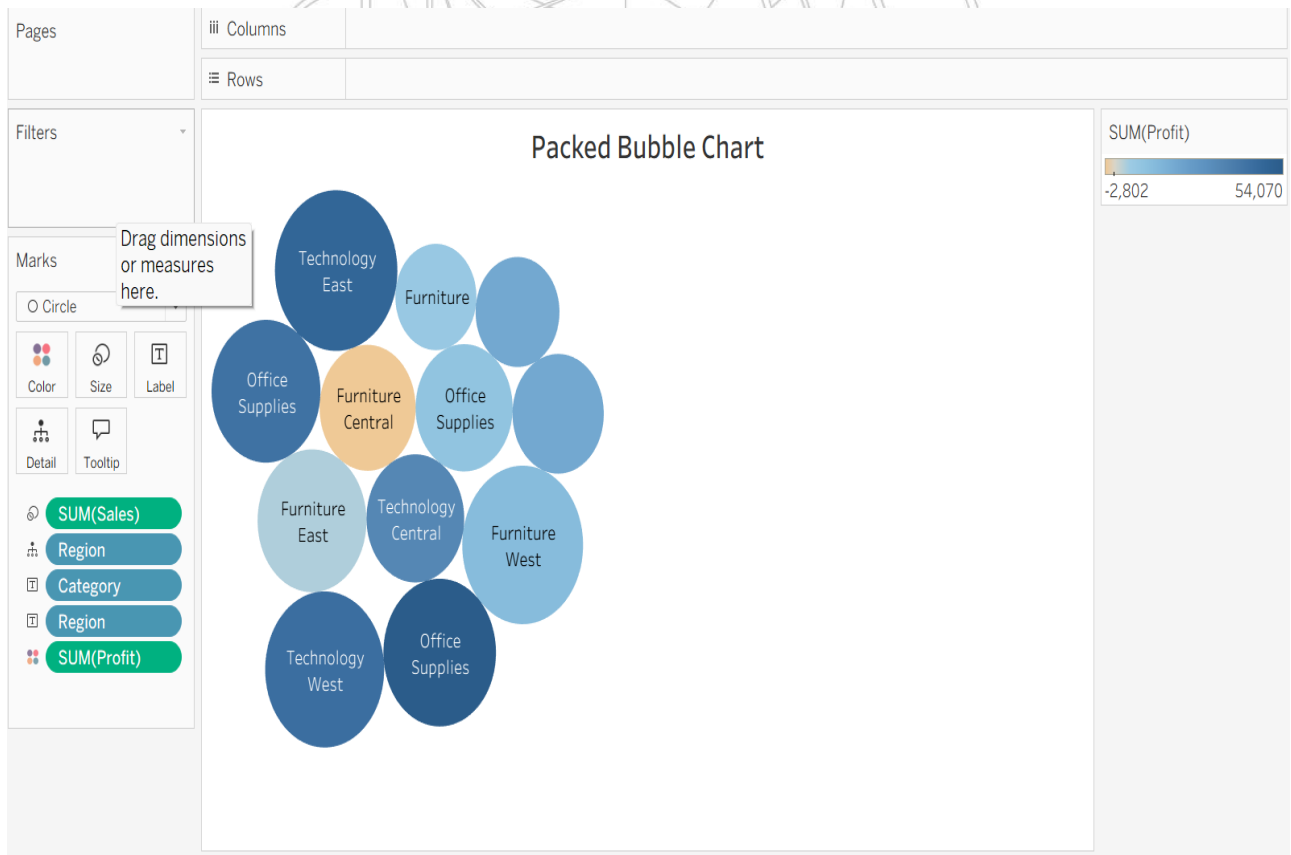
Example:

- Dark blue =High profit
- Light blue =Lower profit
- Orange/Red(If palette changed) = Loss

Add Region to Label

- Drag **region** and drop it to label on **marks card**.
- Now **each bubble** displays **region name**.

Output:



Sheet – 09: Pie Chart

- Drag **profit** and place it on the **columns shelf**.
- Now drag **category** and place it on the **Rows shelf** It creates a **Bar chart** by default.
- Click on **show me** and select the **pie chart**.
- After selecting **pie chart**, Mark the type changes to **pie**.

Sum(profit) moves to:

- Angle
- Size

Category move to

- Color

Now increase Pie chart:

- Select **Entire view**.
- Now, click **size** on marks card move slider right.
- Now the **pie chart** becomes larger.

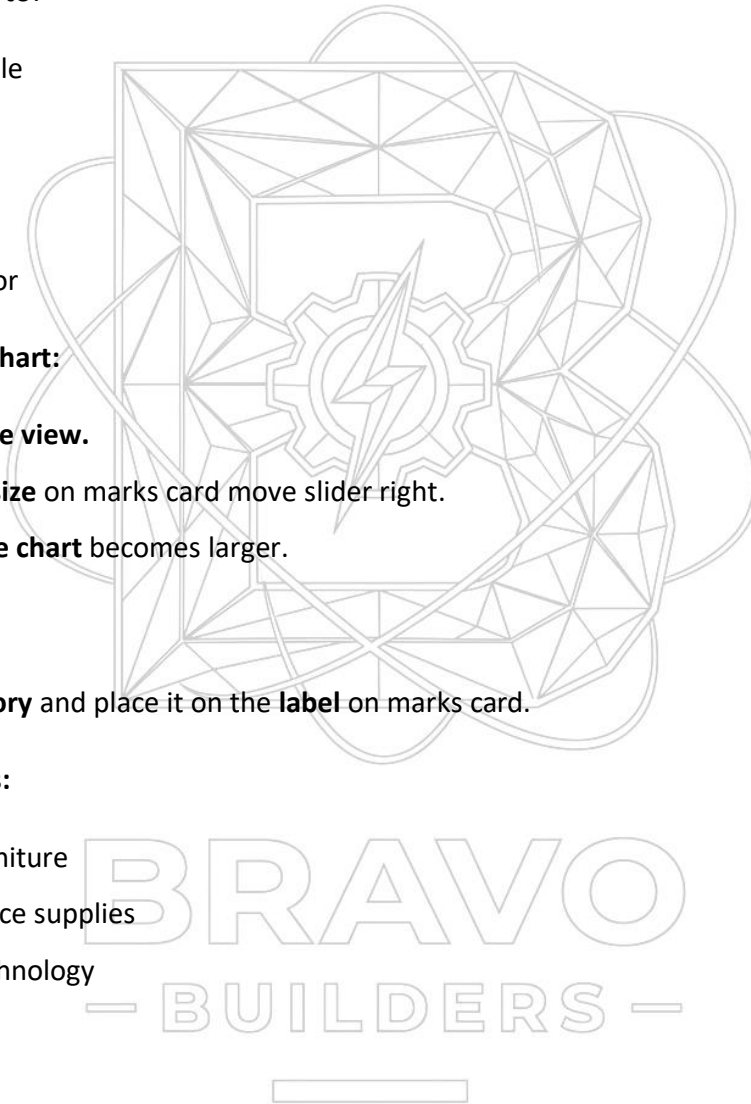
Add labels

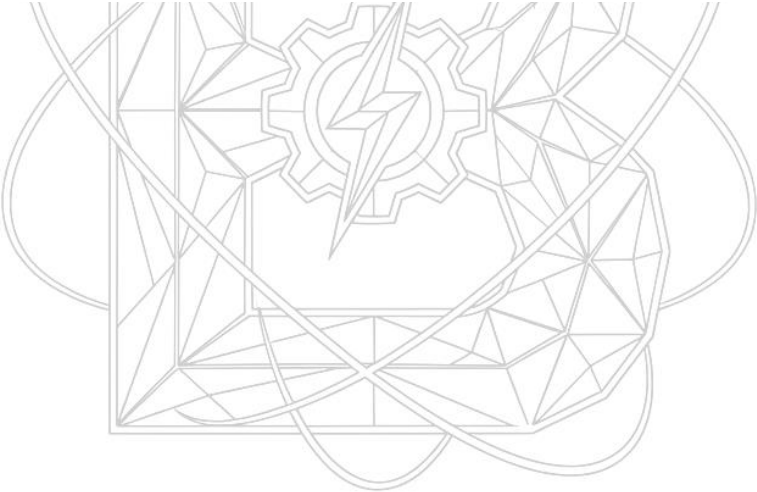
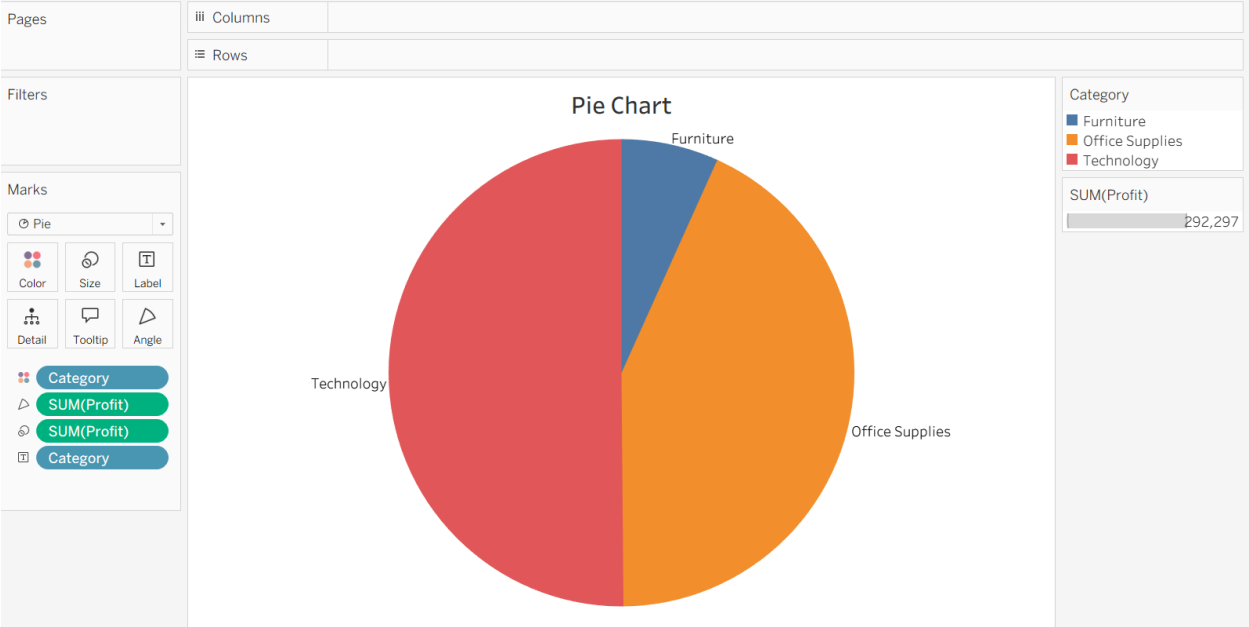
- Drag **category** and place it on the **label** on marks card.

Now chart displays:

- Furniture
- Office supplies
- Technology

OUTPUT:





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Sheet – 10: Scatter Plot

- Drag **sales** and place it on the **columns shelf**.
- Now, tableau creates **horizontal axis**.
- Drag **profit** and place it on the **Rows shelf**.
- Now, tableau creates **vertical axis**.
- Now, you get one **mark scatter plot**.

Add Category to Color

- Drag **category** and place it on the color in **marks card**.

Now marks split into:

- Furniture
 - Office supplies
 - Technology
- Each category gets different color.

Add Region to Detail

- Drag **region** and drop it on **details** in marks card.

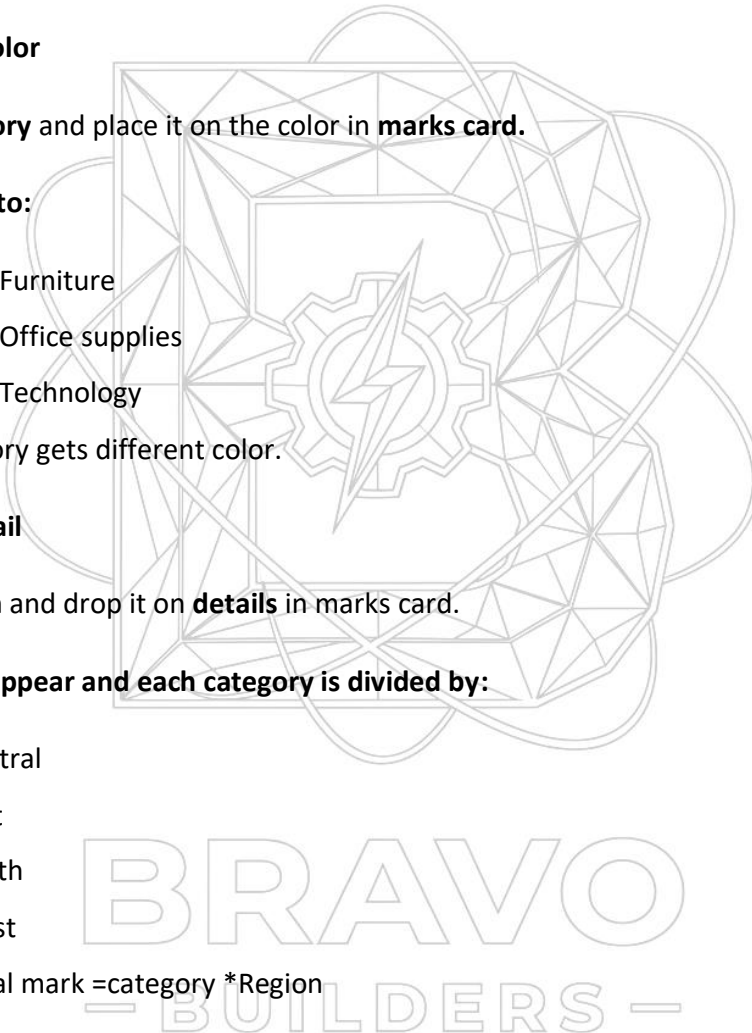
Now more marks appear and each category is divided by:

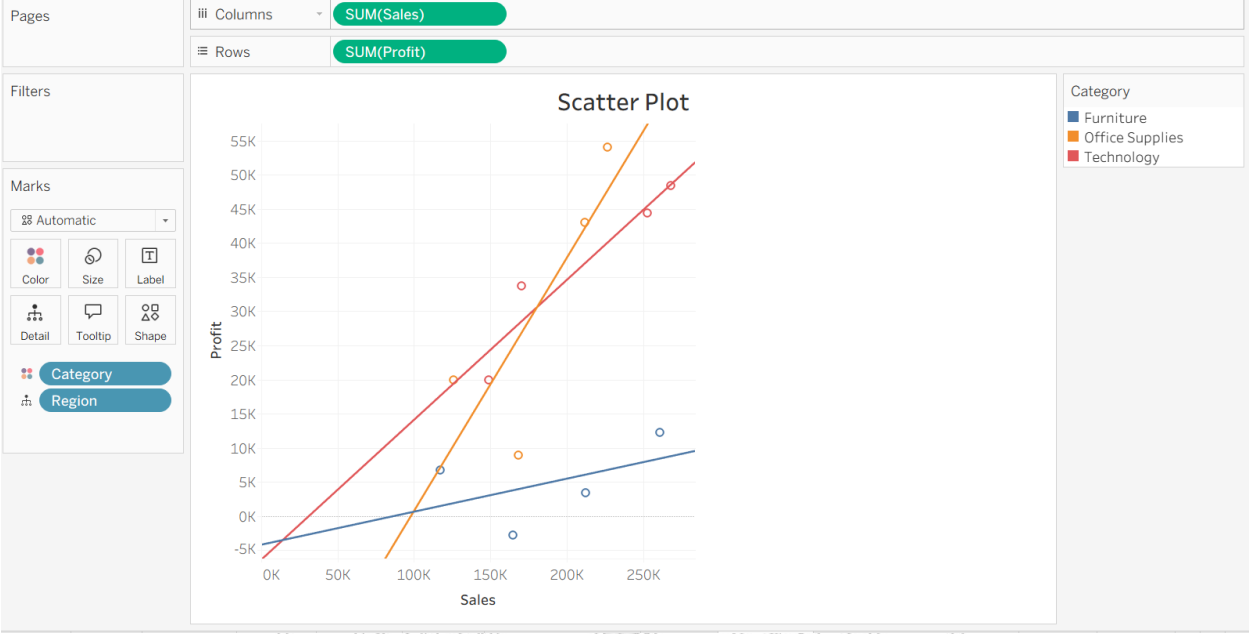
- Central
- East
- South
- West
- Total mark =category *Region

Add Trend Lines

- Now, click **Analytics** Pane and Drag **Trend Line**.
- Now drop it on the **chart** Then, **select: Linear**.

OUTPUT:





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Sheet – 11: Text Table with Percentages

- Drag **Order Date** and place it on the **columns shelf**.

Add Sub-category to Rows

- Drag **Sub-category** and place it on the **Rows shelf**.

Now a table structure is created.

- Rows=sub-categories
- Columns=Years
- Drag **sales** and place it on the **text** in **marks card**.
- Now, each cell shows: **SUM(sales)** For each **year** and **sub-category**.
- Drag **Region**.
- Then **drop** it on the left side of the **sub-category in rows shelf**.

Now rows become:

- *Central--->Chairs
- *Central--->Phones
- *East--->Chairs
- *East--->Phones etc...

Convert Sales to Percentages

- Now, right-click **SUM(Sales)** on marks card.
- Select add **table calculation**.

Configure table calculation in dialog box

- Calculation type: percent of total
- Compute using: Pane (down), Then click ok.

OUTPUT:



Pages

Columns: YEAR(Order Date)

Rows: Region, Sub-Category

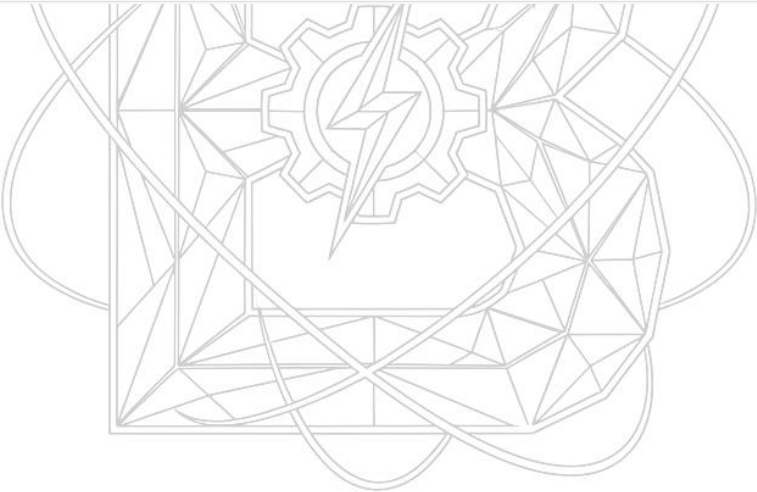
Filters

Marks

- Automatic
- Color
- Size
- Text
- Detail
- Tooltip
- SUM(Sales)

Text Table with Percentages

Region	Sub-Catego..	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.26%	3.45%	3.61%	4.22%
	Phones	9.56%	18.82%	13.39%	15.70%
East	Storage	10.68%	8.10%	8.63%	9.28%
	Supplies	0.42%	0.32%	2.89%	2.98%
	Tables	7.50%	6.67%	9.37%	7.16%
	Accessories	4.47%	11.40%	3.41%	6.85%
	Appliances	4.53%	4.26%	5.22%	5.73%
	Art	1.01%	1.13%	1.11%	1.27%
	Binders	5.40%	9.18%	10.81%	6.84%
	Bookcases	8.02%	12.51%	3.30%	3.42%
	Chairs	16.89%	12.75%	12.70%	14.50%
	Copiers	2.24%	2.15%	12.24%	11.27%



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- BUILDERS -



Sheet – 12: Tree Map

- Drag **Sub-category** and place it on the **columns**.
- Now, you will see **category** names on **horizontal axis**.
- Drag **sales** and place it on the **Rows shelf**.
- Now tableau will create a **bar chart** by default.
- Now click on **show me**.
- Then select the **Treemap** to create rectangles for each **sub-category**.

For example:

- The larger dark box = high sales
- The small light box = low sales
- Drag **ship mode** and place it on **color in marks card**.

Now the rectangles are colored by:

- First class
- Second class
- Standard class
- Same day By the controls size.
- Now click **undo** and remove **ship mode**.
- Then drag **profit** and place it on the color

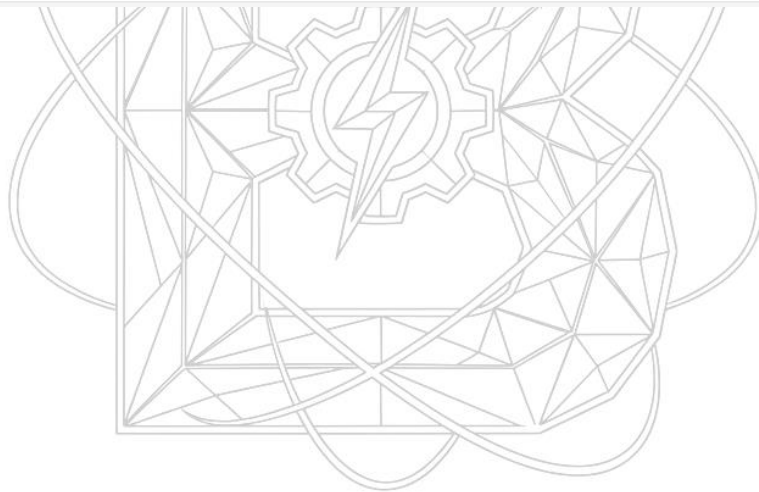
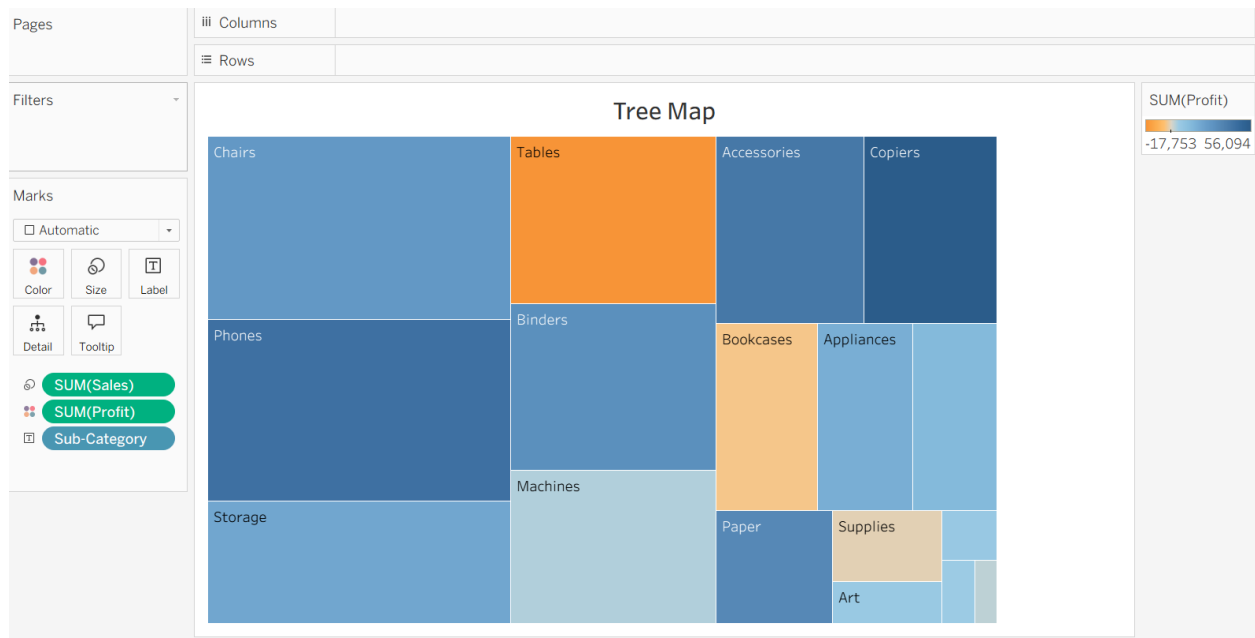
Now will you see

- Size =sales
- Color=profit

OUTPUT:

BRAVO
— BUILDERS —
—





BRAVO

— BUILDERS —



Sheet – 13: Donut Chart

Creating the calculated field as zero

- Now go to **analysis menu** and select **create calculated field**.
- Name the **field as Zero**
- Then add the formula area, Type Formula:0
- **Click okay.**

Zero to Rows

- Drag **zero** to the **rows shelf**.
- Now **zero** will be added in the **measures section**.

Convert to Pie

- Click **size** in the **marks card** and select the **Pie**.
- From the toolbar, choose fit **entire view** to get the clear larger pie.

Create Second Pie for Donut Hole

- Again drag Zero from Measures to the Rows shelf.
- Place it on the right side of the First Sum(Zero).

Create Dual Axis

- Right click on the SUM(zero) in row.
- Then select dual axis. Now both the pie charts overlap.

Create the inner circle

- Click the second SUM(zero), in the **marks card**.
- Click **size** and reduce the size slightly and click **color**.
- Select the **white color** to get the clear **Donut chart**.

Add Region Colors

- Now click the first SUM(zero) on the **marks card**.
- Then drag the **region** to color.

- Each region gets different color.

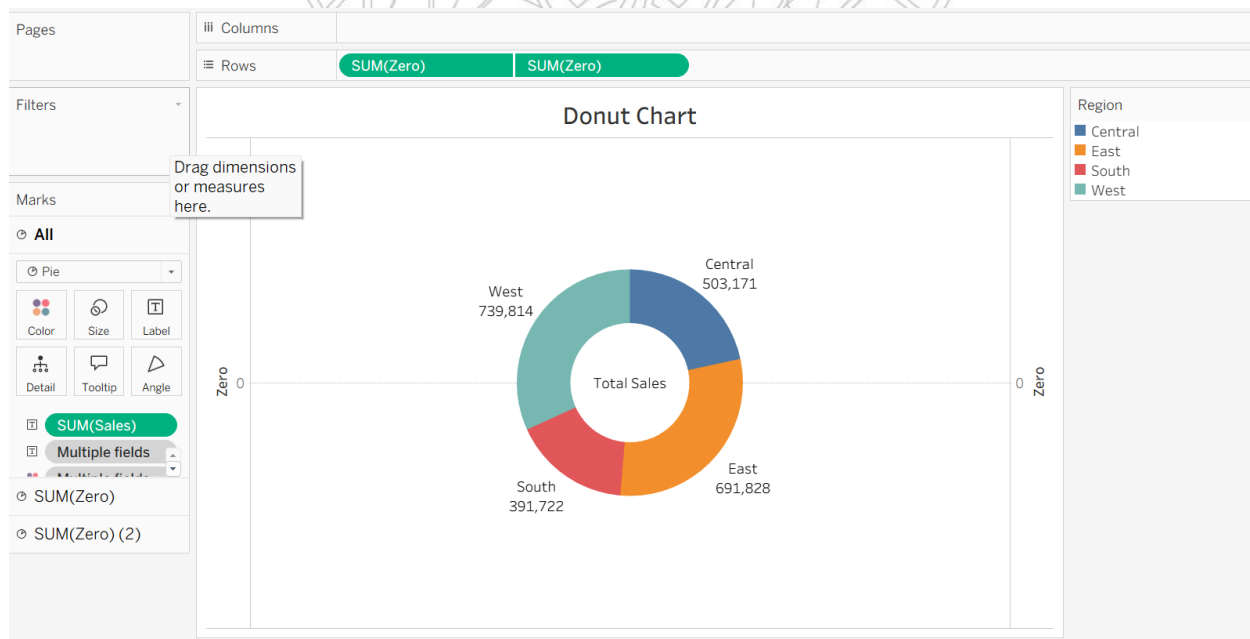
Add labels.

- Drag **region**, **sales** and place it on the label in **marks card**.
- Drag **sales** to Angle.

Add Total Sales in Center

- Click the second **SUM(Zero)** in **marks card**.
- Drag **sales** to the label.
- Click **label Text** and click the **three dots (...)** beside the text editing.
- Type: Total sales, Select **ok**.

OUTPUT:



— BUILDERS —



Sheet – 14: Map Chart

Creating the basic map

- From the **data pane** and Drag **state** to the workspace.
- Then the **tableau automatically generates a map** using latitude and longitude.

Add Sales Information

- Drag **sales** and place it on the color in **marks card**.
- Now the **map** displays sales values with **different color**.

Change Map Type

- Mark card, click the **automatic**.
- Next select map or filled map.
- Shows circles on locations and Filled Map Fills states with colors.

Edit colors:

- Click **color** on the **marks card**.
- Then choose a **color palette (purple)** Click ok.

Add labels

- Drag **state** to label and again drag **state** to label.
- Now each **state** shows its sales value.

Add Region

- Drag **region** to the **filters shelf**.
- Select all region and click ok.

Add Profit Details

- Drag **Profit** to Tooltip.

Format the Map

- Go to Map--->Map layer.
- Adjust:

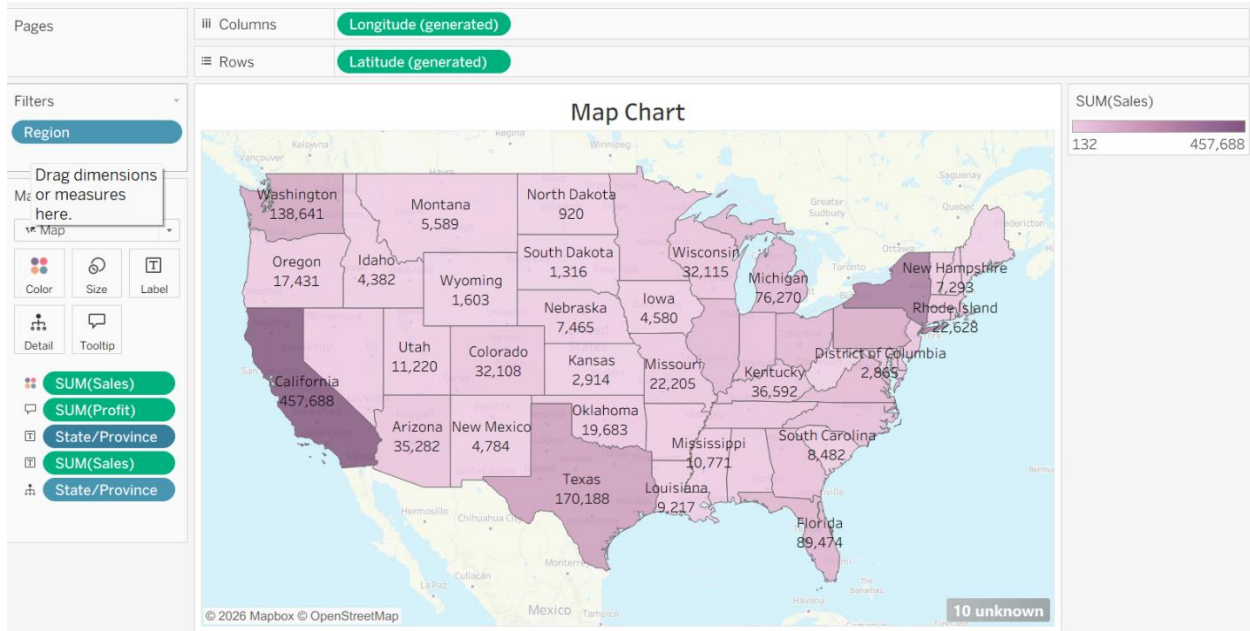


BRAVO
— BUILDERS —



- Background Style(Outdoors)
- Washout(70%)

OUTPUT:



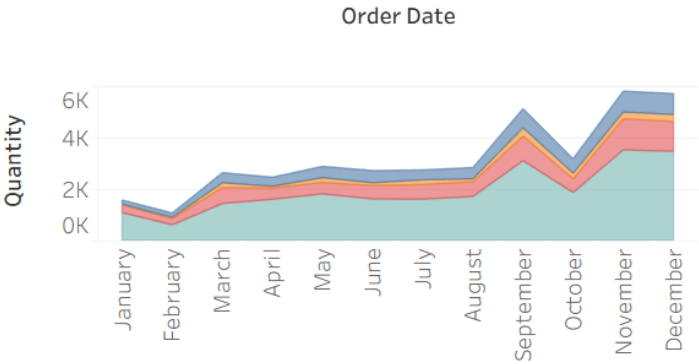
BRAVO
— BUILDERS —



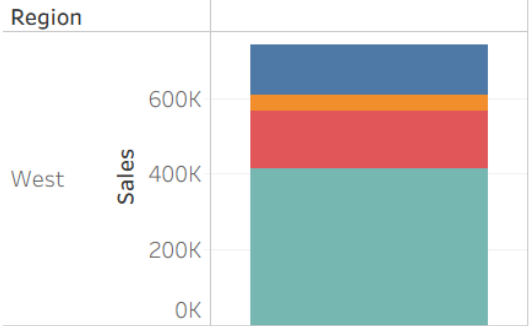
DASHBOARDS

Dasahboard - 01

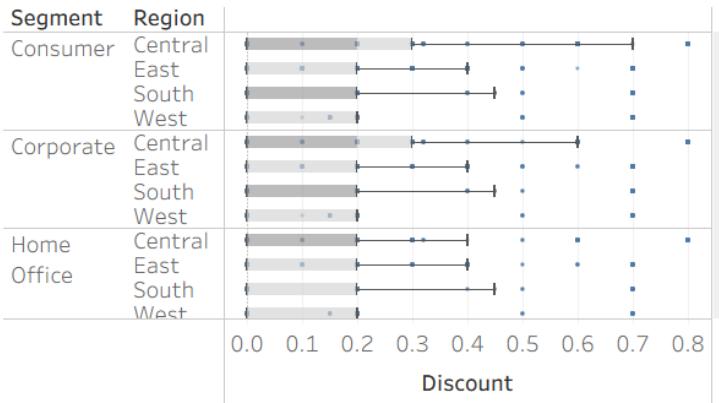
Area Chart



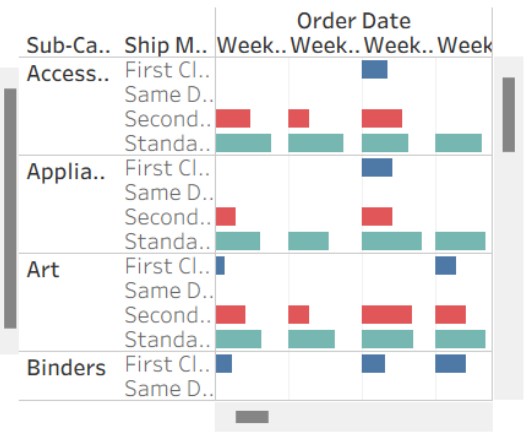
Bar Chart



Box Plot



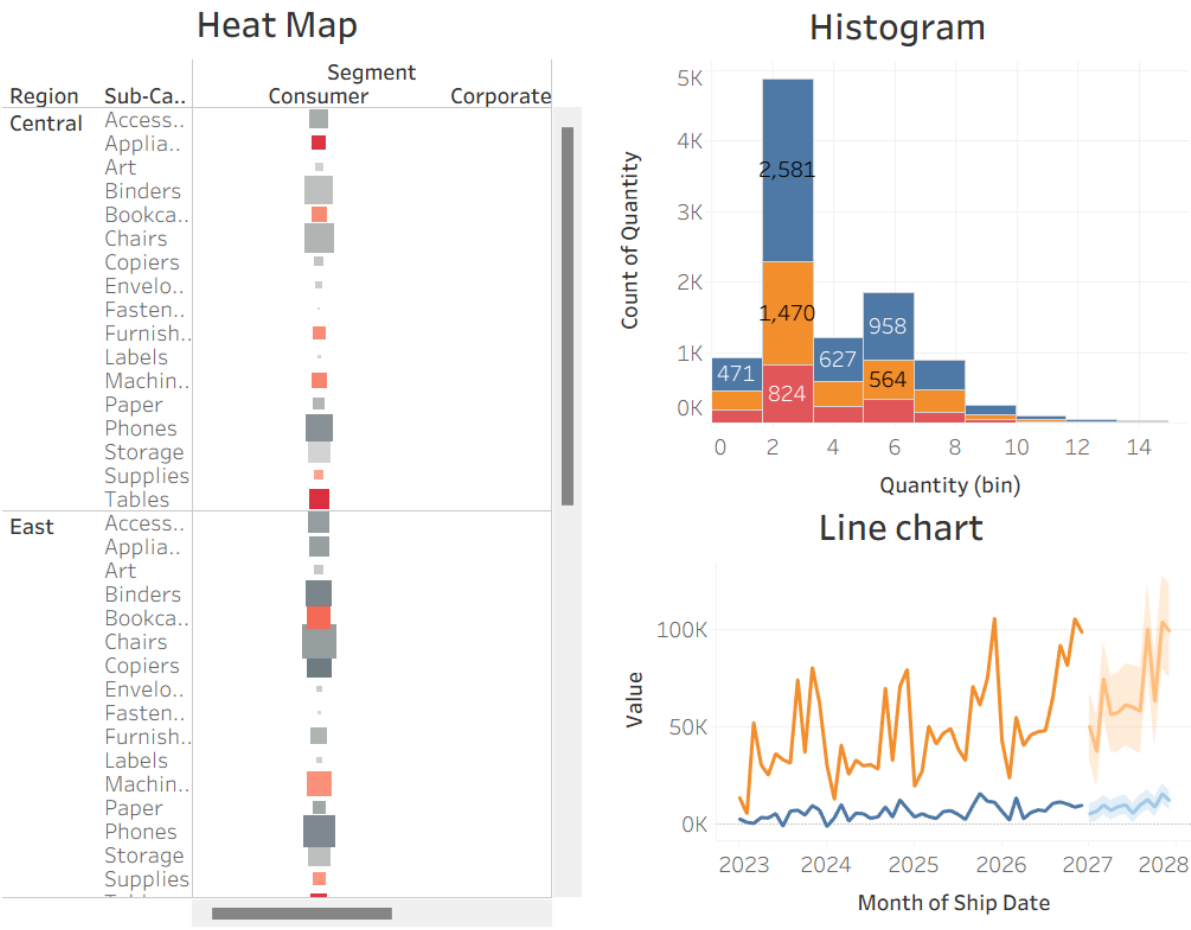
Gantt Chart



BRAVO
- BUILDERS -



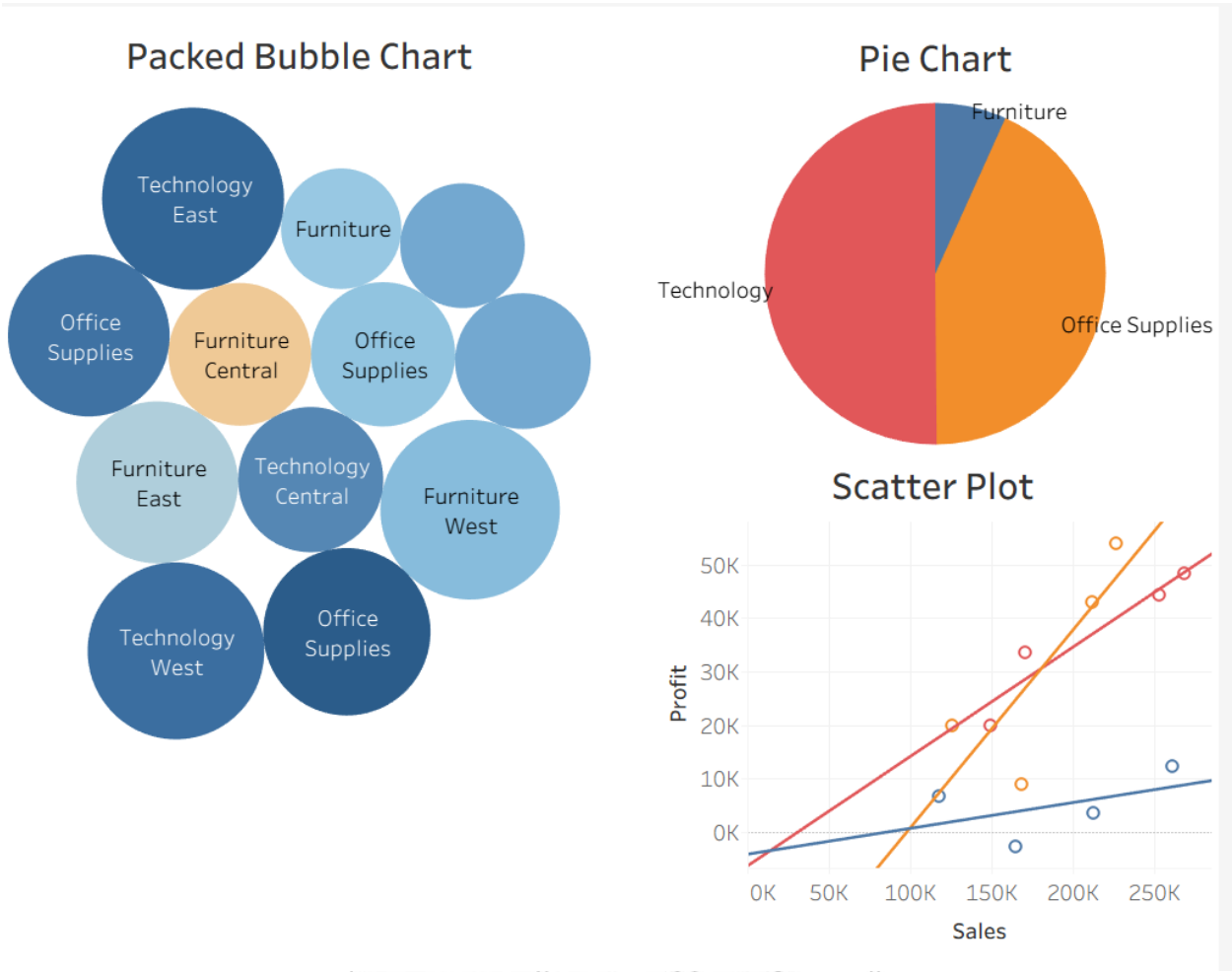
DASHBOARD 2



BRAVO
- BUILDERS -



DASHBOARD 3



BRAVO
- BUILDERS -

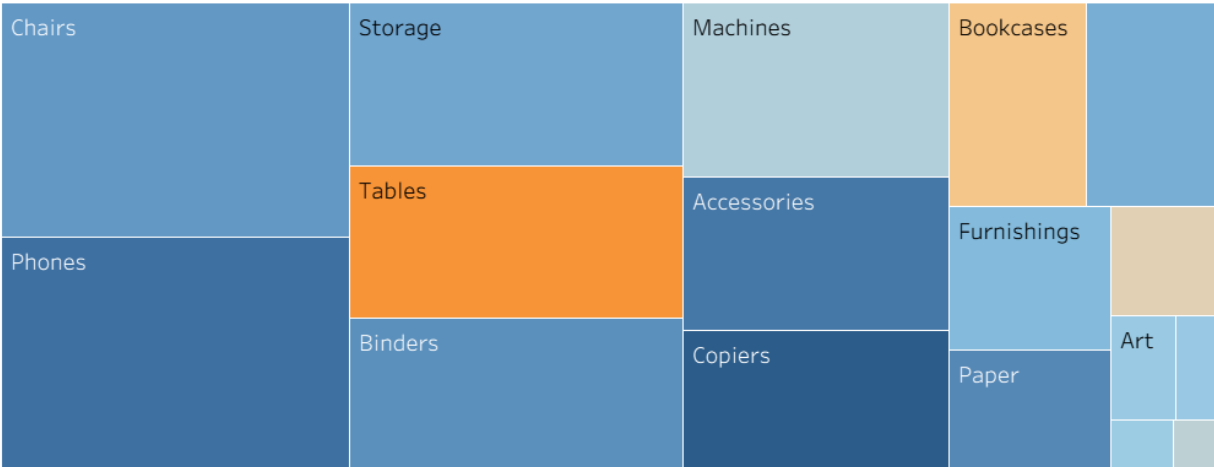


DASHBOARD 4

Text Table with Percentages

Region	Sub-Catego..	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishinas	2.44%	2.46%	3.45%	3.44%

Tree Map

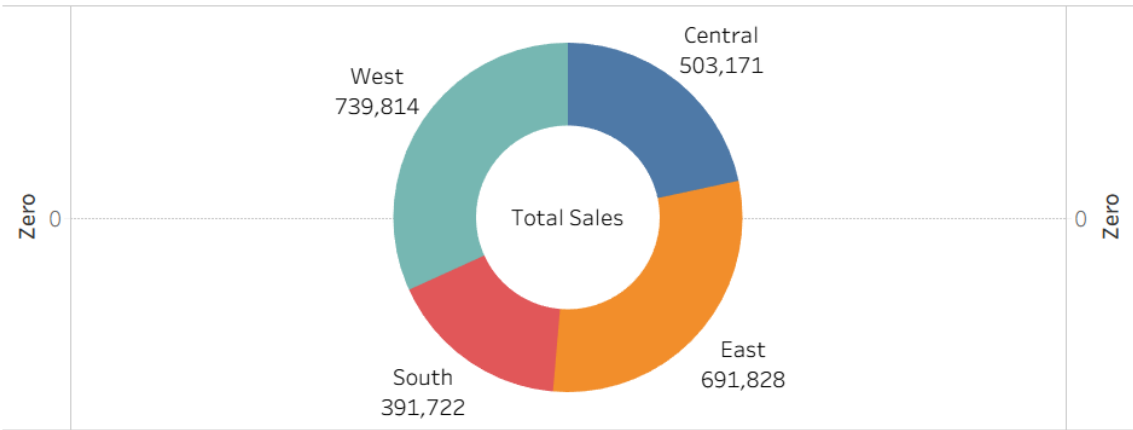


BRAVO
- BUILDERS -

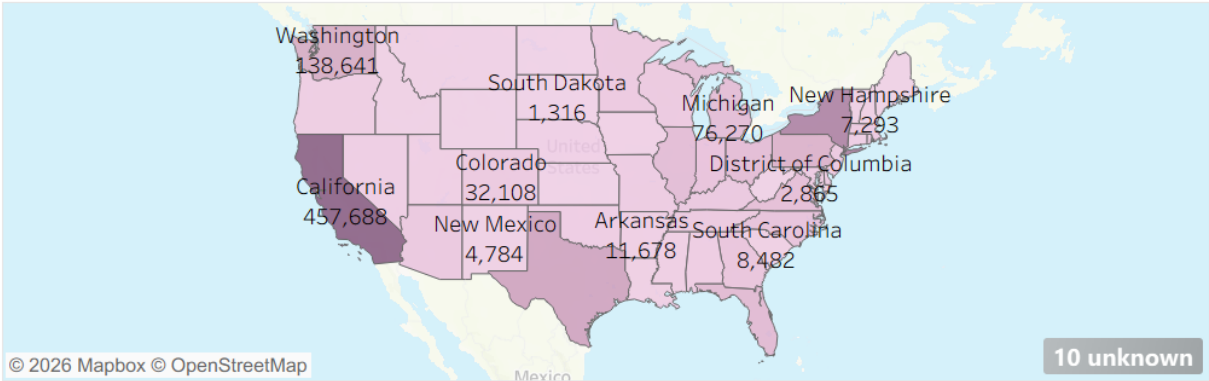


DASHBOARD 5

Donut Chart



Map Chart



BRAVO
- BUILDERS -

